

## End –End Communication at Transport Layer

- a) Implement echo client server using TCP/UDP sockets.
- b) Implement a chat program using socket programming.

**AIM:** To implement (a) an echo client-server using TCP and UDP sockets, and (b) a chat program using socket programming in Python.

### Procedure

1. **TCP echo:** run `tcp_server.py` (binds to 127.0.0.1:5000, listens and accepts a connection), then run `tcp_client.py` in another terminal. Type a message; the server prints it and sends it back. Type `exit` to close the client.
2. **UDP echo:** run `udp_server.py` (port 5001, `SOCK_DGRAM`) and then `udp_client.py`. The client sends each message with `sendto()` and the server echoes it back; `exit` stops both programs.
3. **Chat program:** run `chat_server.py` (port 5002) and then `chat_client.py`. The client sends a message, the server displays it and types a reply, and the two sides take turns until either one sends `exit`.
4. Observe the output in each terminal.

### `tcp_server.py`

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

server.bind(("127.0.0.1", 5000))

server.listen(1)

print("TCP Echo Server started...")

conn, addr = server.accept()

print("Connected:", addr)

while True:

    data = conn.recv(1024)

    if not data:

        break

    print("Client:", data.decode())

    conn.send(data)

conn.close()

server.close()
```

### **tcp\_client.py**

```
import socket

client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

client.connect(("127.0.0.1", 5000))

print("Connected to TCP Echo Server")

while True:

    msg = input("Enter message: ")

    if msg.lower() == "exit":

        break

    client.send(msg.encode())

    data = client.recv(1024)

    print("Server:", data.decode())

client.close()
```

### **udp\_server.py**

```
import socket

server = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

server.bind(("127.0.0.1", 5001))

print("UDP Echo Server started...")

while True:

    data, addr = server.recvfrom(1024)

    print("Client:", data.decode())

    if data.decode().lower() == "exit":

        break

    server.sendto(data, addr)
```

```
server.close()
```

### **udp\_client.py**

```
import socket
```

```
client = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
```

```
print("Connected to UDP Echo Server")
```

```
while True:
```

```
    msg = input("Enter message: ")
```

```
    client.sendto(msg.encode(), ("127.0.0.1", 5001))
```

```
    if msg.lower() == "exit":
```

```
        break
```

```
    data, addr = client.recvfrom(1024)
```

```
    print("Server:", data.decode())
```

```
client.close()
```

### **chat\_server.py**

```
import socket
```

```
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
server.bind(("127.0.0.1", 5002))
```

```
server.listen(1)
```

```
print("Chat Server started...")
```

```
conn, addr = server.accept()
```

```
print("Connected:", addr)
```

```
while True:
```

```
    msg = conn.recv(1024).decode()
```

```
    if msg.lower() == "exit":
```

```
        break
```

```
    print("Client:", msg)
```

```
reply = input("You: ")  
conn.send(reply.encode())
```

```
if reply.lower() == "exit":  
    break
```

```
conn.close()
```

```
server.close()
```

### **chat\_client.py**

```
import socket
```

```
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
```

```
client.connect(("127.0.0.1", 5002))
```

```
print("Connected to Chat Server")
```

```
while True:
```

```
    msg = input("You: ")  
    client.send(msg.encode())
```

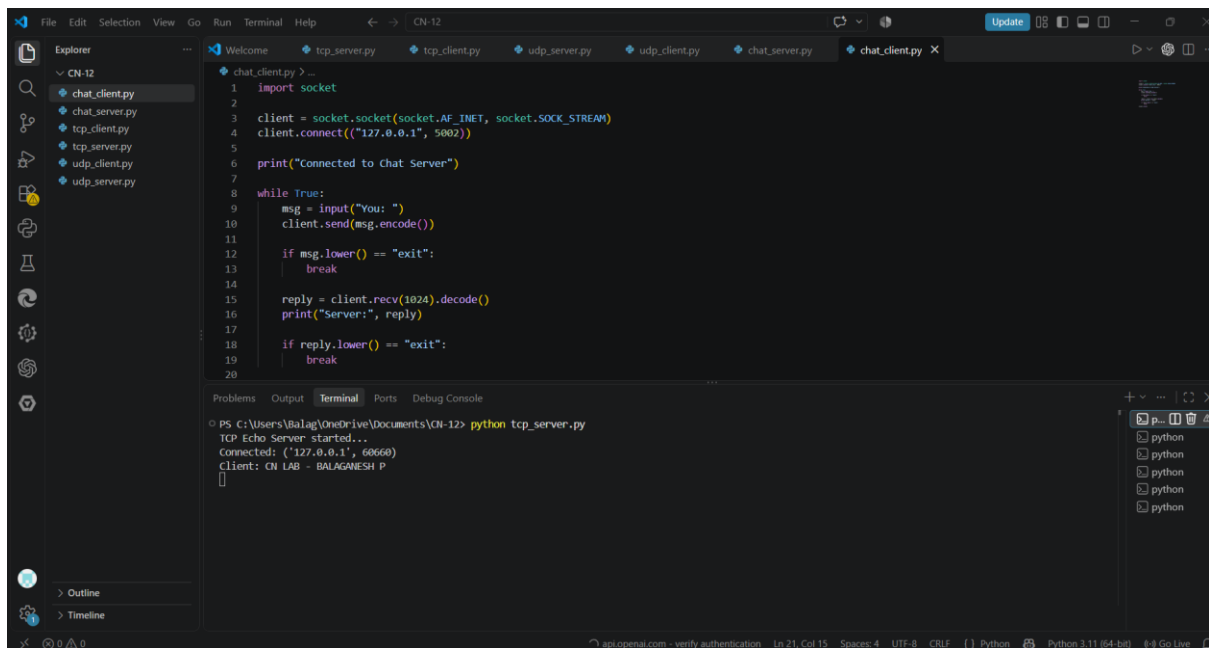
```
    if msg.lower() == "exit":  
        break
```

```
    reply = client.recv(1024).decode()
```

```
    print("Server:", reply)
```

```
    if reply.lower() == "exit":  
        break
```

client.close()



The screenshot shows the VS Code editor with a project named 'CN-12'. The Explorer sidebar on the left lists files: chat\_client.py, chat\_server.py, tcp\_client.py, tcp\_server.py, udp\_client.py, and udp\_server.py. The main editor window displays the code for chat\_client.py, which imports socket, creates a client, connects to '127.0.0.1' on port 5002, and enters a loop to send and receive messages. The Terminal window at the bottom shows the command 'python tcp\_server.py' being executed, with output indicating the TCP Echo Server is started and a client connection from 'CN LAB - BALAGANESH P' is received.

```
1 import socket
2
3 client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
4 client.connect(("127.0.0.1", 5002))
5
6 print("Connected to Chat Server")
7
8 while True:
9     msg = input("You: ")
10    client.send(msg.encode())
11
12    if msg.lower() == "exit":
13        break
14
15    reply = client.recv(1024).decode()
16    print("Server:", reply)
17
18    if reply.lower() == "exit":
19        break
20
```

Problems Output Terminal Ports Debug Console

PS C:\Users\Balag\OneDrive\Documents\CN-12> python tcp\_server.py  
TCP Echo Server started...  
Connected: ('127.0.0.1', 60660)  
Client: CN LAB - BALAGANESH P  
█

Problems Output Terminal Ports Debug Console

PS C:\Users\Balag\OneDrive\Documents\CN-12> python tcp\_client.py  
Connected to TCP Echo Server  
Enter message: CN LAB - BALAGANESH P  
Server: CN LAB - BALAGANESH P  
Enter message: █

Problems Output Terminal Ports Debug Console

PS C:\Users\Balag\OneDrive\Documents\CN-12> python udp\_server.py  
UDP Echo Server started...  
Client: CN LAB EXP  
█

PS C:\Users\Balag\OneDrive\Documents\CN-12> python udp\_client.py  
Connected to UDP Echo Server  
Enter message: CN LAB EXP  
Server: CN LAB EXP  
Enter message: █

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python chat_server.py
Chat Server started...
Connected: ('127.0.0.1', 58511)
Client: HII
You: GOOD MORNING
Client: WELCOME
You: 
```

Problems Output **Terminal** Ports Debug Console

```
PS C:\Users\Balag\OneDrive\Documents\CN-12> python chat_client.py
Connected to Chat Server
You: HII
Server: GOOD MORNING
You: WELCOME

```

## Result

The echo client-server was implemented using both TCP and UDP sockets, and a chat program was implemented using socket programming in Python.

In the TCP test the message "CN LAB - BALAGANESH P" was echoed by the server, in the UDP test "CN LAB EXP" was echoed back, and in the chat program the client and server exchanged messages (HII, GOOD MORNING, WELCOME) successfully.